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A review on fatigue damage mechanism in hydro turbines

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ABSTRACT

Well knowing a good-running hydraulic turbine has important operational and financial benefits to those who operate a plant. Fatigue damage is the most fundamental failure type of hydraulic turbines. Existing flaws due to fatigue and their risk limit the operating time of a unit. Some plants have been temporarily shut down for up to a month and sometimes longer to repair these fatigue-induced damage, which has resulted in enormous economic losses. Fatigue problems must be solved or effectively prevented to ensure that turbine units run safely and steadily within their design life. This paper reviews loading features and some key issues (e.g. different load operations, start-up, emergency shut-down, load rejections, and runaway) on the fatigue damage, and provides the latest information about different prediction approaches. At last, it also attempts to present an overview of the complete failure modes, therefore other types of failure including cavitation, erosion and ingested bodies are introduced briefly.
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1. Introduction

Hydropower is a renewable energy source based on the natural water cycle, captures the kinetic energy of falling water that comes from a reservoir, a river, and waterfalls. Over the past century, hydropower has become a proven, extremely flexible, and well-advanced technology. The world's installed hydropower capacity in 2009 was 926–980 GW. By 2011, hydropower is the largest renewable energy in the electricity sector, and it contributed 17% of worldwide electricity supply and over 72% of the world's renewable electricity. China, United States, Brazil and Canada are the countries

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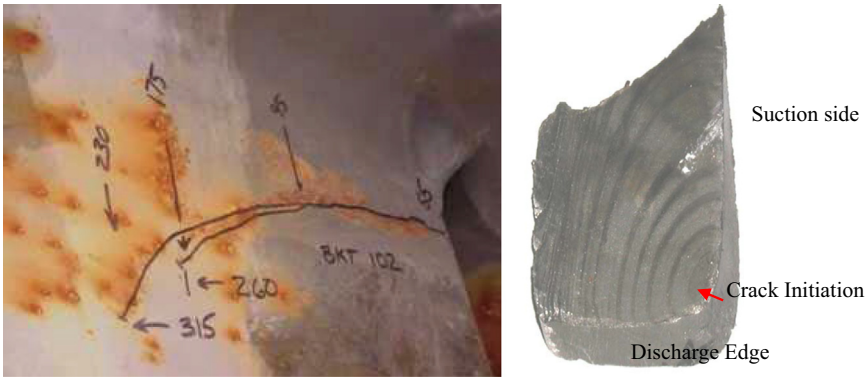


Fig. 1. Crack fracture surface.

Table 1
Failure information in some hydropower plants.

Plants	Country	Turbine type	Rated Unit Capacity (MW)	Failure location	Failure time	Shutdown (month)	Loss (million $)
Sayano-Shushenskaya	Russia	Francis	640	Bolts in #2, the plant destroyed	2009	Under repaired	13,000[a]
DaChaoShan	China	Francis	225	Runner	2001	4	63.71[b]
G.M. Shrum	Canada	Francis	261	Runner and wicket Gates	2002	14	258.67[b]
Porjus second	Sweden	Francis	240	Draft tube	2000	2.5	42.47[b]
XiaoLangDi	China	Francis	330	Runner	2001	4	93.44[b]
ErTan	China	Francis	561	Runner	2000	1	39.71[b]
ShuiKou	China	Kaplan	200	Piston rod	2006	1	14.16[b]
Khimti	Nepal	Pelton	12	Needle and buckets	2003	1	0.85[b]

[a] Direct economic loss from the official accident report. In fact, the indirect pecuniary loss is much large than this value.
[b] No data available in references. An approximate estimation based on 80% of effective generating period times unit capacity times residential retail price (12.29 cents/kWh) in the USA at February 2015. Maintenance costs are not included.

which have the largest hydropower generation capacity [1,2]. While current hydropower technology is very mature, there is still some room for further improvement. During operation, the acute interaction of turbine components occurs frequently and there are potentially severe consequences. For a majority of hydropower plants, turbines have operated for decades, and in many of these hydropower plants the operating conditions have changed significantly from the original conditions. These changes in operating conditions can lead to excessive vibration: in some cases, fatigue cracks have formed in parts of the turbines [3–5]. A typical blade crack due to fatigue cyclic loads is shown in Fig. 1. The formation and propagation of cracks may still lead to premature failure of or damage to key turbine components. Existing flaws and their risk limit the operating time of a unit. Some plants have been temporarily shut down for up to a month and sometimes longer to repair these fatigue-induced damage, which has resulted in enormous economic losses, as listed in Table 1 [6–12]. Compared with hydraulic performance issues (i.e. output, efficiency and cavitation), it is more difficult to assess the durability of the equipment [13]. Because of the aforementioned, fatigue problems must be solved or effectively prevented to ensure that turbine units run safely and steadily within their design life. In this paper, an overview of the fatigue damage, the most important failure mechanism of hydro turbines, will be presented in details, and other types of failure mechanisms such as cavitation, erosion, and ingested bodies will also be presented briefly.

2. Fatigue failure

There are different stages of fatigue damage where defects may nucleate in an initially undamaged section and propagate in a stable manner until catastrophic fracture occurs. For this most general situation, the progression of fatigue damage can be broadly classified into the following stages: (i) structural and microstructural changes which cause nucleation of permanent damage; (ii) the creation of microscopic cracks; (iii) the growth and coalescence of microscopic flaws to form 'dominant' cracks; (iv) stable propagation of the dominant macrocrack; (v) structural instability or complete fracture. The conditions for the nucleation of microdefects and the rate of advance of the dominant fatigue crack are strongly influenced by a range of mechanical, microstructural and environmental factors [14,15]. Fig. 2(a) shows a part of the crown detached from the runner while the machine was in operation. The detached part passed through the machine, causing further damage. Fig. 2(b) shows a close-up of the broken part of the runner. The analysis of the part revealed a fatigue problem. Beach marks can easily be identified in the crack, which was propagated from the T-joint between the runner blade and the crown [16].

Life estimation of turbine components require numerous intrinsic mechanical factors and extrinsic factors. Fig. 3 schematically summarizes these factors and their-relations [17]. These factors act in synergy to establish the turbine's static strength, cavitation resistance, erosion resistance, corrosion resistance, impact resistance, fracture toughness and/or fatigue resistance. But difficulties exist still in the design stage, model tests and even in-situ measurements. One tricky issue is to find out a proper fatigue criterion for hydro turbines. The general physical model is simple geometric structure under axial-loading. Contrarily, the hydraulic loading are complex and multi-axial. In general, failure behavior in hydro turbines is difficult to evaluate by experimental tests or is experimentally inaccessible. Because it needs to test a large number of materials and structural components in a very short time. The current work is to develop the general failure physical model by introducing the correctional parameters based the existing measurement data of hydro turbines. Another tricky issue is how to predict prototype loadings based on the model tests. It is related to a popular focus on similarity relationship between model and prototype machines. At present, lots of work have been carried out to establish the hydraulic similarity law through model test and prototype test. Even so, there is a debate in the extrapolation of model test results to prototype values [18].

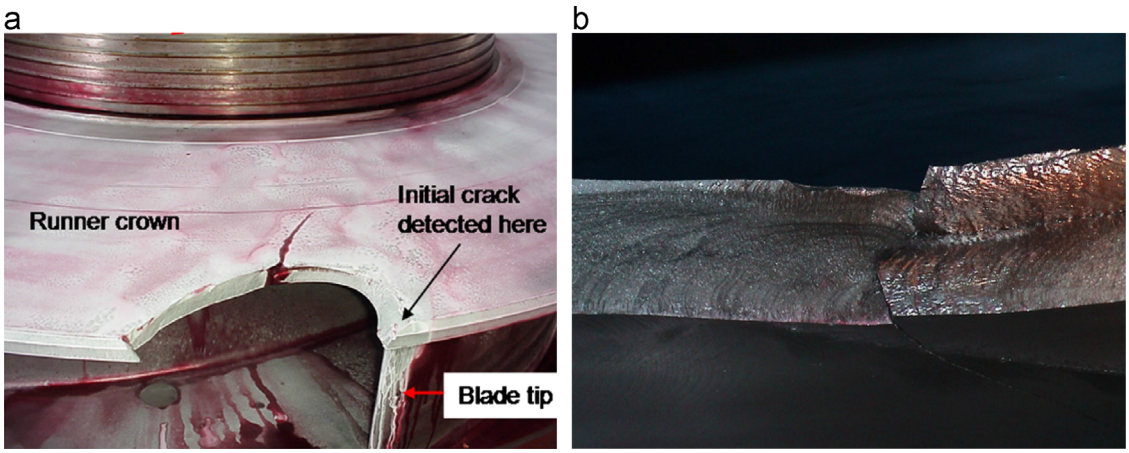


Fig. 2. (a) A picture of the broken runner. (b) View of the crack on the runner with the beach marks.

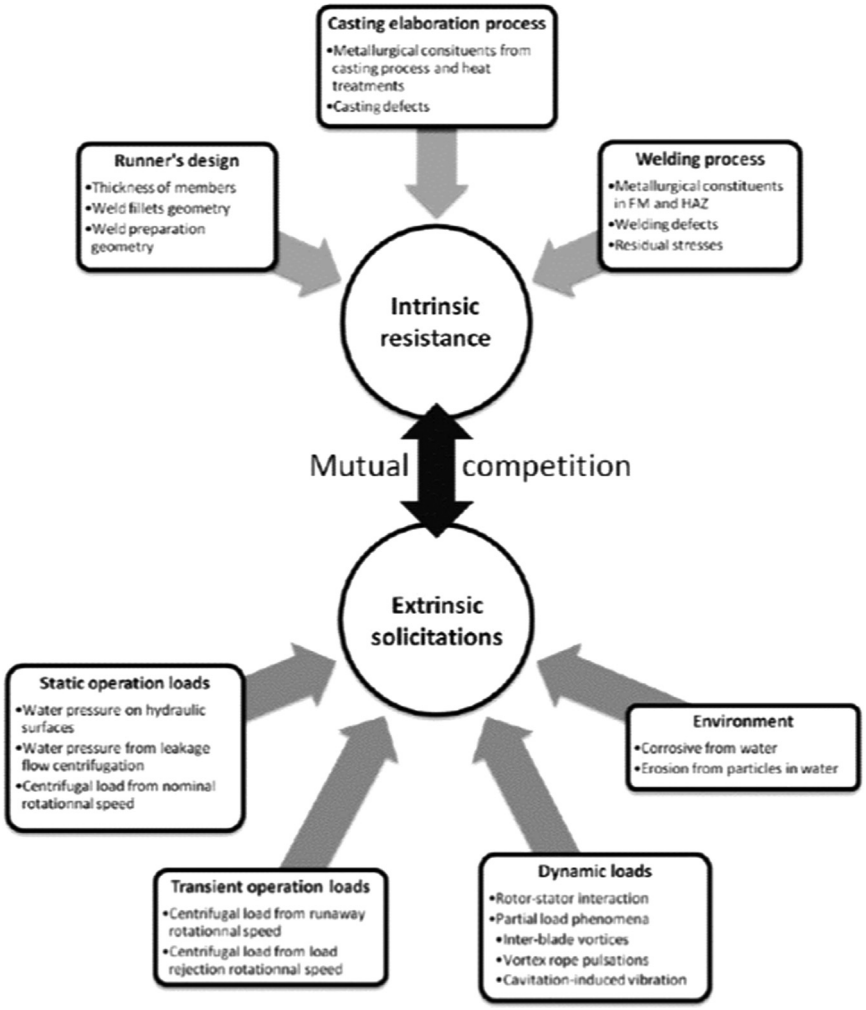


Fig. 3. Schematic representation of mutual competition between the intrinsic mechanical resistance of hydro turbine runners and the extrinsic solicitations to which they are subjected.

Because too many factors can lead to the prototype real value different from those calculated by scaling formulas. The essential reason is that all of losses lead to efficiency will change in prototype machine. Very few structure stress measurements were carried out in model tests because the runner is rotating under a narrow flow passage, let alone the prototype in the field. This also limits the development of the life estimation of hydro turbines.

2.1. *Loading features in operating conditions*

Not only fatigue strength, but also the performance of the pressure fluctuations, noise of the key positions, vibrations and swings of the key positions, sediment abrasion, cavitation and turbine efficiency, have closely connections with actual operating situations of the power plants [19,20], see Fig. 4. Brekke [21] discussed the turbine

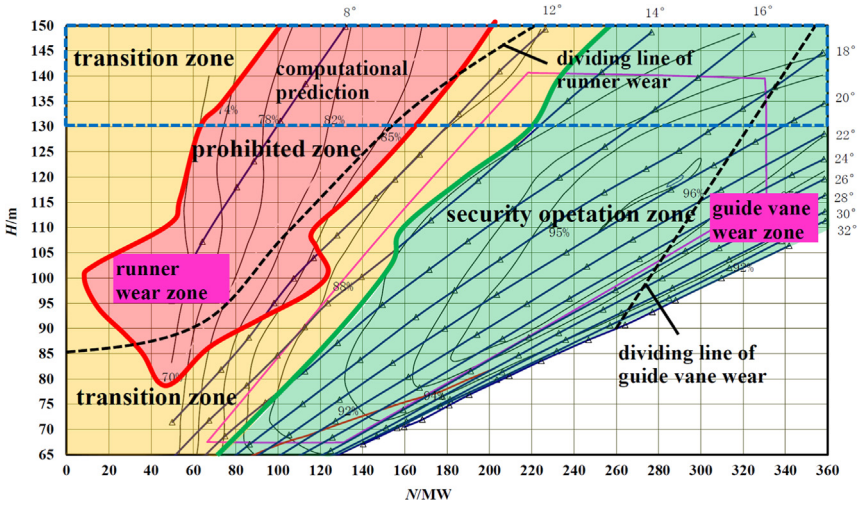


Fig. 4. District map of unit operation.

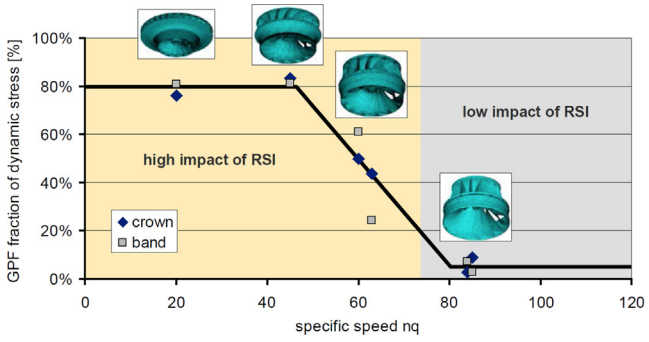


Fig. 5. The ratio of the stress amplitude at GPF to the entire dynamic stress in terms of peak-to-peak/2 values is shown in Fig. 10 versus the specific speed *nq*.

design for safe operation. He pointed that resonance with the natural frequencies of the runner have not been proven to be the reason for the problems of noise and or blade cracking of low specific speed high head Francis turbines. In high specific speed Francis runners there are normally not observed similar dominating high frequency pressure pulsations from the blade passing as for high head runners. However, resonant problems and stress amplitudes driven by low frequency draft tube problems have been observed. The reason for the absence or weak influence from the blade passing frequency is the larger distance from the guide vane outlets to the runner blade inlets at the crown compared to the band. The natural frequency of the low head runners may also in some cases be in resonance with the low frequency pressure surges in the draft tube. For example, when Francis turbine is operating at partial load region, the draft tube surges-so-called “Rheingans oscillations” [22] – due to interaction between hydraulic excitation sources (i.e., vortex rope precession) and Eigen-frequencies are expected to induce high frequency loading [23]. This phenomenon generates strong vibrations and noise that may produce fatigue failures on the mechanical elements of the machine. The ratio of the stress amplitude at guidevane passing frequency (GPF) to the entire dynamic stress in terms of peak-to-peak/2 values is shown in Fig. 5 versus the specific speed n_q.

Turbine's operation schemes, mainly through start–stop sequences, load rejection and runaway events as well as power variations, can cause variations of static and transient operation loads [17,24,25]. The load cycles thus incurred are considered as low-frequency high-intensity (LFHI) fatigue cycles, or low-cycle fatigue (LCF) cycles. They are associated with large stress variations occurring but at very low frequencies ($\sim 10^{-8}$–10^{-4} Hz). However for most operating conditions, dynamic loads are associated with small stress fluctuations occurring at high frequencies ($\sim$10–200 Hz). Examples of typical phenomena that induce dynamic stresses in runners include rotor–stator interactions [26,27], vortex rope pulsations [28], inter-blade [29] and Von Karman vortices [30]. In the event of a correspondence between the runner's natural frequencies and exciting frequencies related to these phenomena, mechanical resonance can amplify these stress fluctuations, potentially leading to a high fatigue damage rate and significant crack propagation [31]. Fig. 6 shows an example of resonance fatigue damage due to Von Karman vortices at the XiaoLangDi power plant in China [10]. One of possible solutions is the modification to the blade discharge edge. The load cycles induced by dynamic loads can thus be referred to as high-frequency low-intensity (HFLI) fatigue cycles, or high-cycle fatigue (HCF) cycles. Many researchers [32,33] continuously investigate all kinds of dynamic loading and the impact on the fatigue life of hydroelectric components. Especially off-design conditions and transient events of Francis runners were addressed in some recent research activities. The understanding of dynamic loading has been highly improved, through numerous strain gauge measurements at prototype runners [34], model test [35], and CFD or fluid–structure interaction (FSI) analysis [28,36,37].

High frequency pressure fluctuations will always be created because the runner blade must pass the wakes behind the guide vanes. The frequency of these blade passing pressure shocks is the runner speed multiplied by the number of runner blades at the runner inlet i.e. including the splitter blades, if the runner is furnished with splitter blades. However, inside the runner in the runner channels, the frequency of the pressure shocks will be the speed of the runner multiplied by the number of guide vanes. In some cases too high stress amplitudes have caused fatigue cracking of runner blades. The reason for this has normally been the shape and thickness of the blades versus the magnitudes of the pressure pulsations [21]. In the rotating reference frame of the runner, the entire excitation pressure field is varying in space and time and may be described as a linear combination of rotating pressure mode shapes given by [36,38–40]

$$p(t,\vartheta) = \hat{p}\cdot \cos\left(n\cdot Zg\cdot 2\pi\cdot f_N\cdot t - k\cdot\vartheta + \varphi\right) \quad \text{with } k = \pm m\cdot Zr - n\cdot Zg \text{ and } m,n = 0,1,2,\ldots \tag{1}$$

where ϑ is the angle coordinate in the rotating frame of reference. The frequency of the runner rotation is given by f_N, but the runner is excited by the gate passing frequency GPF$=Z_g f_N$ and its higher

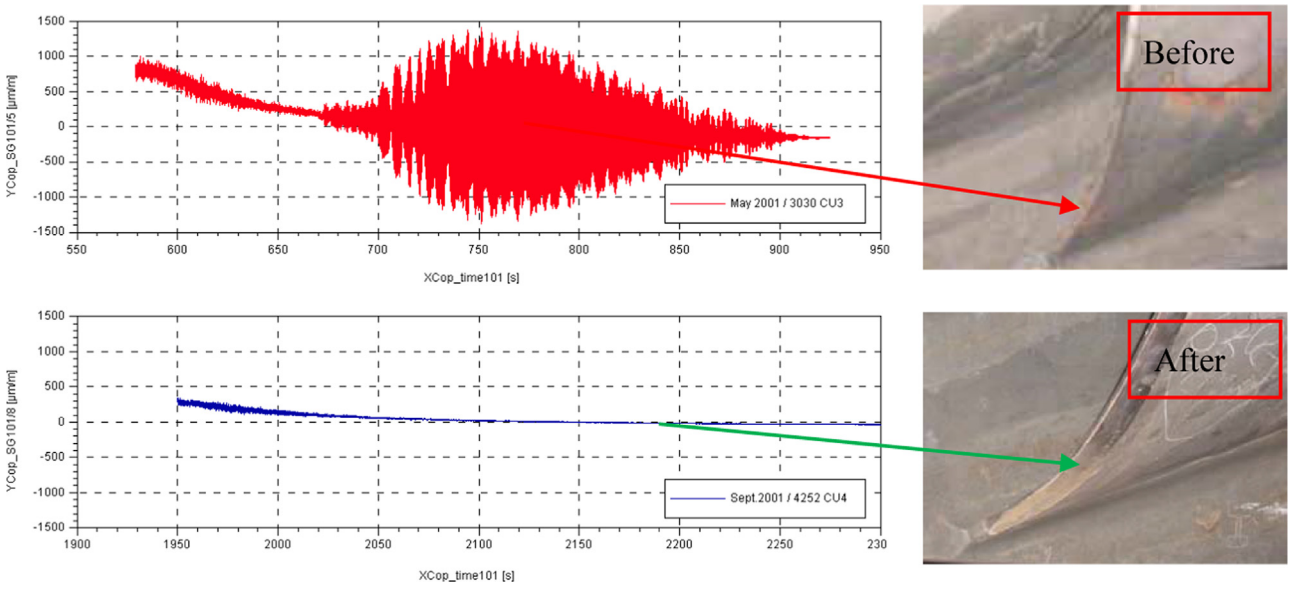


Fig. 6. Blade strain vs. time during unit coast down after wicket gate closure.

harmonics. For a given combination of Z_r runner blades and Z_g guide vanes, mode shapes (characterized by the k-number) and corresponding frequencies are defined by the integers m and n. Amplitudes $\widehat{p}$ and phase angles φ of all appearing pressure mode shapes can be identified from unsteady CFD analyses by means of Fourier series expansions in space and time.

2.2. *Effect of unsteady flow in stationary events*

In hydro turbines, unstable flow behavior is a key contributing factor to pressure fluctuations. It can lead to unwanted noise, vibrations and material fatigue issues. This instability will erode and fatigue the material of the turbine by the continuous impulsion released by the collapsing cavitation bubbles and the swirling vortex [41,42]. Due to the instability of highly swirling flow, vortex rope – concentrated vortical structure with rotational frequency of about 20–25% of the runner rotation (low frequency pulsations), arises which appears downstream of the turbine runner. Vortex rope will result in so called draft tube surge which propagates into whole hydraulic system eventually leading to fatigue cracks of the blades or power swing phenomena of the electrical generator [43].

Seidel et al. [33] and Aschenbrenner et al. [44] summarized the flow physics at different operating conditions with focus on cavitation phenomena of Francis turbines. At high load, when the fluid enters the draft tube, it tends to towards the machine axis causing a swirl against the runner rotation. Usually, this operating condition is stable with small fluctuations in the draft tube. However, Flemming et al. [45] found under certain circumstances, the vortex core volume may fluctuate and interact with the whole power plant system such that unstable pressure oscillations flow. Most of the pressure fluctuations were measured through experiments. The results showed that pressure fluctuations between runner and guide vanes were mainly caused by rotor stator interaction [46], which may become a dominant factor at high load and around best efficiency point [36]. At part load, the outflow will lead to a backflow in the center of the draft tube cone and a vortex rope of helical shape. In the meantime, because of low pressure, the vortex will form a cavitation bubble in the draft tube, and this vortex rope will create periodic pressure fluctuations in the runner at a low frequency which is lower than (about 20–25%) the rotational speed of the turbine [33]. These components of draft tube pressure pulsation (DTPP) may be separated because they differ by their spatial relationships and by their propagation mechanism. In high and part load regions, they can be distinguished as so-called synchronous and asynchronous pulsations [47]. At low part load, less regular pulsations are always present. The statistical method [47,48] permits to separate the stochastic (random) component from the two traditional 'regular' components. The flow in the runner still tends to larger radii than that in part load. Moreover, secondary flow effects between runner blades will occur and cause channel vortices with low pressure regions in the vortex core which generate cavitation. In this operating region, the flow induces high amplitude pressure fluctuations of stochastic nature with a broad banded frequency distribution which decrease from very low to medium frequencies. Although a number of investigations into both full load and part load operating conditions is reported, the research about deep part load condition has not been well mentioned yet. And as expected, extreme operating conditions, as deep part load, are the most demanding for the turbine structure. Pressure measurements on the runner blade to investigate the fatigue life of the runner were conducted in the low discharge condition (about 30% of the nominal discharge) by Farhat and Lowys et al. [49,50]. Yamamoto et al. [51] unveiled and investigated cavitation phenomena observed at deep part load condition by high speed visualizations and pressure fluctuation measurements. Magnoli and Maiwald investigated the stability and pressure pulsations in Francis turbines at overload, part load and deep part load based on numerical simulations and experimental model test results [52].

Until recently, operation at very low part load was regarded as something exceptional and unattractive, due to the associated low hydraulic efficiency. However, especially in Europe, because of massive expansion of renewable energies (e.g. wind and solar energy), hydroelectric installations become the most efficient way in the interest of electrical grid stabilization, energy storage and peak load supply. In consequence, a trend exists to operate hydro turbines more frequently at low part load and for extended times. In these turbines, some field tests including runner strain measurement showed the high dynamic stresses measured [24] caused by the part load vortex at low load operation are even in the same order of magnitude as at speed-no-load (SNL). The operation in this zone for a reduced time period will not lead to significant damage to the runner. However, if the runner is running for an extended period of time in the low load region, the

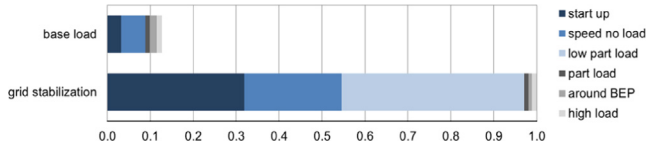


Fig. 7. Relative damage contributions of the load universes.

lifetime of the runner may be consumed quickly [53]. The relative partial damage contributions, shown in Fig. 7, are derived from strain gauge measurements at a prototype runner during operation [33].

An example with more drastic consequences has been observed when a unit in a remotely controlled powerhouse experienced the closure of four wicket gates due to shear pin failures. The unit was not equipped with protective devices to detect shear pin failure and operation continued. Due to the strongly unbalanced flow, the runner was pushed toward the low pressure side of the spiral case and strong contact with the bottom ring occurred at the runner band seal. At the end, the unit failed with major damages to the runner and the wicket gates requiring extensive repairs [54].

Asymmetry of casing is another important factor that could cause severe unsteady flow. The flow contour of spiral or semi-spiral casings, in particular those in older machines, is sometimes not well-designed and may not produce a symmetric in- or out-flow for the runner. Seen from the runner, the asymmetric boundary condition translates to a periodical disturbance through which the runner rotates. The flow and pressure at a given region of the runner, for instance the exterior edge of every blade, is faced with an alternating load that may in extreme cases be high enough to cause fatigue problems [13]. Asymmetric flow in the spiral case due to inadequate spiral case design may have some effects on the mechanical behavior of the runner. Usually, this effect would not be noticeable. However, fatigue analysis would have to account for this effect together with the effect of the guide vane wakes.

2.3. *Effect of transient events and no-load (NL) conditions*

Transient events are referred as harmful conditions for hydro turbines [55]. They will often lead to high amplitude pressure fluctuations on different parts of the turbine, including the rotating parts, which will affect a turbine residual-life [56]. During transients, Francis turbine experiences cyclic stresses, asymmetric forces on the runner, and wear and tear, all of which reduce the operating life of the components [57]. Hence, transient events often account for most of the damages sustained by hydro turbines during operation. They affect the turbine lifetime both by accelerating crack propagation on the runner blades [58] and damaging the bearings [55]. Trivedi and Cervantes Michel gave a very good review on effect of transients on Francis turbine runner life [57]. Most notable of No-Load operating conditions are speed-no-load, runaway and transients such as load rejections. NL operation is typically highly dynamic as well. Pressures and strains strongly fluctuate under NL. During transient events the turbine can temporarily pass through a NL condition [59].

During load variation, the guide vanes experience unsteady pressure differences on pressure and suction sides, the flow condition worsens and becomes unstable in the runner blade passages which are mainly responsible for runner oscillations and asymmetric loading on the blades, and the vortex instability increases [60,61]. Under the worst-case conditions, FSI can lead to structural failure of the runner blades. In recent years, turbine manufacturers have encountered some serious problems of crack in the runner blade due to asymmetric dynamic forces at the runner inlet [62].

Some of the potentially most damaging continuous operating conditions for hydro turbines are the NL conditions. The stochastic pressure loads under no-load operating conditions are significant based on structural fatigue considerations [63]. NL operating conditions can occur in the life time of a hydro turbine, either at machine start, during load rejection or in exceptional cases as a result of ramping or sluicing operation. At NL conditions the flow passes through the turbine without power generation, but with non-negligible flow rate, and therefore all the potential energy in the flow has to be dissipated. This takes place through a mechanism where the runner channels are partially pumping, thus generating large scale unsteady vortex structures which, by their nature, break down into smaller and smaller vortices until energy dissipation occurs at the smallest scales [59]. The mean contribution of speed-no-load to fatigue life of Francis runners can be significant [64]. However, pressures and strains strongly fluctuate under NL operation. The unsteadiness of this condition with its peak-to-peak pressure and strain variations is relevant for fatigue life in [65]. Some recent studies on conditions around NL have been performed in the field of reversible pump turbines, e.g. [61,66].

During start-up and shutdown of the turbine, the runner accelerates and decelerates in an unknown manner or according to the servomotor stroke. Low cycle fatigue transient events induced due to unfavorable acceleration/deceleration can cause damage to the runner blades, and greatly influence turbine's life expectancy [67,68]. Compared with other load cases, the start-up contribute a lot to the damage of each runner. In terms of actual damage, the damage caused by one start-up equals the damages of many hours, even many days of low load operation or SNL [69]. And One start-up reduces the runner lifetime as much as many years of operation under normal operating conditions [67]. Torque fluctuations during start-up slow down the synchronization process and are undesirable during peak-hour demand. This behavior also delays start-up of the turbine. There is practically no load on the turbine shaft, and the turbine operates close to runaway speed, which is more critical [61]. The generator may sometimes fail to synchronize due to a mismatch of the phase angle between the power grid and the generator output terminal ($\sim$120° or $\sim$180°). The failure results in rapid fluctuations of torque and inlet pressure head, and in extreme cases, the rotating components of a generator and turbine may collapse [70]. The runner startup has been identified has one of the most damaging operating conditions. While shutdown does not lead to severe damage of the unit if it is carried out normally. This is because that the turbine is first brought to low load in a gentle manner before it is disconnected in a normal shutdown, and thus the rough NL condition is passed at a low level of speed and flow only. By contrast, there is no way to avoid the pulsations during the start [13]. Nicolet et al. [70] found emergency shutdown of a pump-turbine led to excessive loading on the wicket gates and severer oscillations in the head, discharge, speed and torque, resulting in shearing of the safety pin. The energy dissipated in terms of pressure wave and remaining absorbed by runner. Therefore the abnormal shutdown of the turbine can damage the rotating structure and the connected operating instruments.

Load rejection is one of the most severe transient operating events experienced by a turbine. During load rejection, a turbine inevitably passes through conditions of zero and negative torque, at elevated speed. High stochastic pulsation of pressure and mechanical stresses combines with elevated centrifugal load. In pump-turbines and Francis turbines with very low specific speed, the machine may even temporarily enter a condition of reverse pumping. Through experiments, Liu et al. found that pressure fluctuations in vaneless space and torque fluctuations in the runner initially increased with the rise in speed and then were reduced by the closing of the wicket gates. In a total load rejection situation, rapid pressure fluctuations were captured in the turbine passages, leading to runner torque fluctuations after the load rejection. The torque fluctuations remain until the runaway speed [71]. Gagnon and Leonard [72] investigated runner blades deformation of two hydropower plants. The study

showed that in both cases the most fatigue damage occurs during load rejection.

Runaway, although this is a rare occurrence, is the most damaging event that can happen to a hydro turbine. Running the turbine at its runaway speed is mandatory test at the turbine commissioning. So far, few studies have focused the flow dynamics during runaway [56,71,73,74].

Load variation, start–stop, and total load rejection cannot be stopped by the plant operators, but related damages can be minimized with appropriate strategies. Proper selection of operating scheme may have a significant improvement to the runner life. For example, analysis and study of crack propagation by Gagnon et al. [58] showed that significant improvement in runner life is possible by using improved start–stop schemes rather than predefined schemes. To allow for increased reliability, runner life, and safe operation, turbine manufacturers and academic researchers are required to focus on investigating more about the dynamic behavior of hydro turbines under transient conditions. Attention should be given to the estimation of the magnitude of forces in terms of unsteady pressure pulsations in stationary and rotating components of turbines [57].

3. Metallographic and materials investigation of fatigue failure

The improvements in numerical tools and the continuous demand for better turbine performances are pushing the materials to their limits. In other words, fatigue analysis is asking for a better understanding of materials behavior [75], and defects and inspection to avoid blade cracking [76].

Welded joints of turbine runners are one of the most critical parts of Francis turbines due to the presence of welding discontinuity and high stress. For example, the fracture of the Francis turbine runner blade appeared some months after the second weld repair realized on site on fatigue cracks initiated near to the trailing edge at the junction with the crown [15], see Fig. 8. Because of thermal cycles, solidification, cooling distortion and residual stresses, welded joints always include discontinuities of different types and sizes. It will always be a certain danger for fatigue cracking growth started from minor material defects exceeding a certain limit in welds and also in base material in the joints between blades and crown or band in the blade channels in a Francis runner [76]. Some specific parameters will limit welding flaw dimensions in some or all direction based on the joint geometry, material and welding procedure. If discontinuities of critical size remain undetected, fatigue cracks might initiate and propagate in these zones because of dynamic in-service stresses leading to high repair costs and long down times [77]. Fatigue failures are more likely to occur at the trailing edge of the runner welded joints mainly as a result of the combined effect of dynamic hydraulic load fluctuations and material discontinuities. High residual stresses in the welds caused blade cracking by the pressure pulsation load in the blade channels, even if these pulsations were moderate [52]. Furthermore, the welded joints are likely to contain flaws which may act as favorable sites for fatigue crack propagation. Habibzadeh-Boukani et al. [77] discussed common welding flaws, their dimensions and orientation, as well as their probability of occurrence in the runner joints.

Up to 1972 the Francis runners were produced by 16% Cr 5% Ni steel which is a Martensitic steel with some small amount of Austenite. However some problems came up even if the electrodes used should be identical to the material composition in the runners. The problem was that weld composite was pure Austenite by normal cooling to 50 °C where the change from Austenite to Martensite should occur for the given composition of Cr/Ni steel. The reason for this deviation was detected because Nitrogen (Ni) was added to the mantel of the electrodes in order to give a more flexible Austenite weld deposit without danger of cracks in the

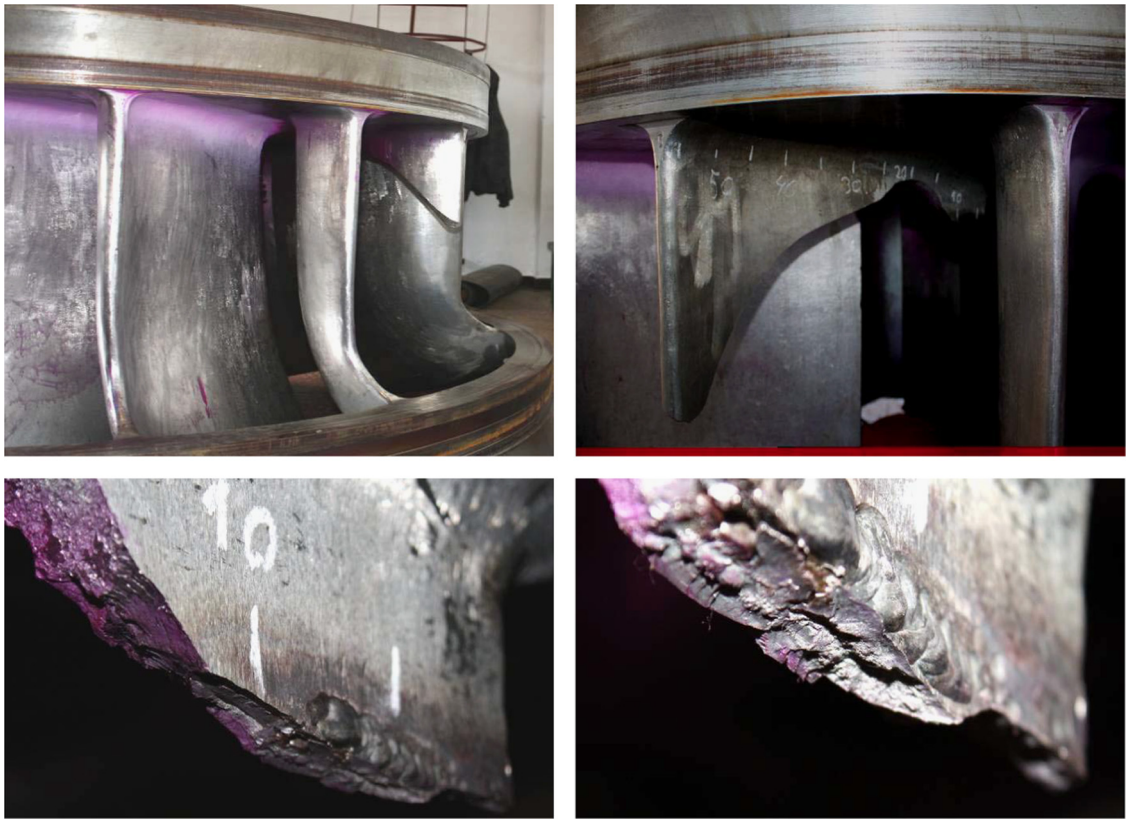


Fig. 8. Detailed views of the welding seam in the vicinity of the runner crown.

weld during cooling after stress relieving up to 580 °C with following cooling to a Martensitic homogeneous deposit at room temperature [76,78]. Welding of new runners after 1978 was made with the new method and both welds and base material were uniform in runners produced by Kvaerner Brug [79]. Hot cracking rarely happens in the martensitic stainless steel runners due to solidification of the weld metal in delta-ferrite phase [80]; ferrite solubility is high for sulphur and phosphorus so that it does not allow low-melting compounds to form in the weld metal.

4. Numerical methods to evaluate fatigue damage and predict remaining life

Generally speaking, in order to optimize maintenance and operation of units and to limit the cumulative damage, electrical utilities have to understand the full impact of start/stop cycles. However, reproducing these transient events numerically is still a challenge due to the lack of general guidelines and the complexity of the physics involved [81].

4.1. Traditional and common cumulative fatigue damage approaches

In order to assess the ability of a turbine to operate under a certain condition before fatigue damage, the load universe describing how the hydropower plant will be operated and component stresses due to static and dynamic loading have to be predicted by the designer and manufacturer. Direct two way coupling methods are very efficient to simulate stresses in the runner but they are very expensive in computation time [82]. Unidirectional numerical method, which enables optimization of each component of the turbine separately and efficiently and provides model updating at each stage to taken into account experimental measurements, is necessary to be developed. Thanks to advances in computing capabilities, it is now possible to realize this method [27]. Firstly a reasonable prediction method for the pressure loads is required. A CFD based method for the prediction of the unsteady pressure loads is most widely used. And secondly a FEM based approach is used for the conversion of the pressure loads into dynamic stresses. Fig. 9 shows an overview of the runner damage factor calculation, which needs the static and dynamic stress analyses [69]. The static stress is calculated by FEM with the pressure loads from steady CFD simulation, which is validated using strain gauges measurements in lots of cases. With last decade developments, the dynamic pressure CFD calculation and the harmonic response of the runner have been carried out frequently in the industry as well [64]. Finally, load cycles can be counted and identified using Rainflow counting. The damage of each load cycle then is evaluated based on *S–N* curve of the material which includes corrections of mean stress, water environment, and surface quality. The entire damage results from linear damage accumulation according to Miner's rule. Fig. 10 describes such a procedure of runner safety check [83]. It should be noted that, structural damping and hydrodynamic damping effect is an essential key factor to determine the amplitudes of dynamic loading. However damping is an inherent factor that cannot be solved by FEM directly. At present, few investigations have been performed to evaluate the effect of flowing water on damping due to FSI for hydro turbine applications [84–86]. More accurate dynamic loads can be obtained by strain gauge measurement instead CFD. Based on measured time histories of strains, the partial damage contributions of single operating conditions as well as the total damage contribution of an assumed load universe can be determined. From correlations of measurement and simulation, some standard procedures, which do not only consider the runner type but also operating requirement of individual projects, have

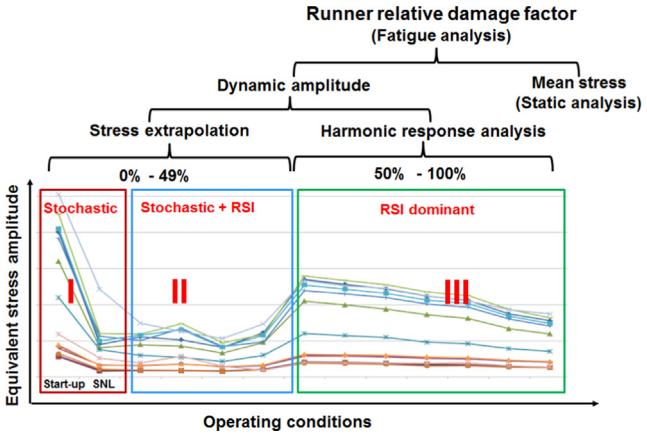


Fig. 9. Overview of the runner damage factor calculation.

been developed to enhance the fatigue life of turbine runners [33,69]. But nonetheless, the experimental method cost much money and a longer time period, so the numerical simulation still got rapidly development in recent years, especially in phase of design or optimization.

Beside SNL and transient events, the origin vibrations and main fatigue contributor for high head Francis turbines in traditional plants was the rotor–stator interaction (RSI), which nowadays can be evaluated quite accurately by means of numerical simulations [36,87].

4.2. Damage tolerance approach (DAT)

Stress-life and strain-life methods focus on the stress and strain state of the structure, which strongly depends on the selected empirical parameters and consequently leads to unreliable estimates. Meanwhile, fatigue life prediction approaches like SN make it difficult to define inspection intervals or repair periods because these model only provide one of two states: undamaged or damaged. In reality, many plant operators and decision makers are more interested in the extent of damage, rather than whether or not there is damage, because such information is useful for planning corrective actions [58]. More advanced approaches, such as a damage-tolerance approach, may be more appropriate for assessing existing structures. This approach to engineering design considers a damage tolerance that is based on the assumption that flaws can exist within every structure and will propagate with usage. Using these types of approaches, the remaining life of a component can be estimated as the time to reach a critical crack length. Because crack growth models are more complex than time to failure approaches, it is obvious that a crack growth period should be known very well for a reliability demonstration. Nevertheless, these different approaches can be combined to maximize the available information of the life of a structure [88]. For damage tolerance approach, the characteristics of remaining flaws in the runner or other parts of turbine significantly affect the life prediction. Therefore, developing a more reliable damage tolerance approach would decrease the risk of cracking and provide a better characterization of the discontinuities in terms of their size [77]. Hydro-Québec used this method to evaluate the fatigue life based on the propagation of existing flaws [89]. Liu et al. [90] developed a damage tolerance approach based on linear elastic fracture mechanics (LEFM) to describe crack behavior using concepts from both applied mechanics and material science by combining the effects of stress states and material flaws. This new method has been successfully applied into the piston rod fracture estimation of a Kaplan turbine installed in a hydropower plant in China.

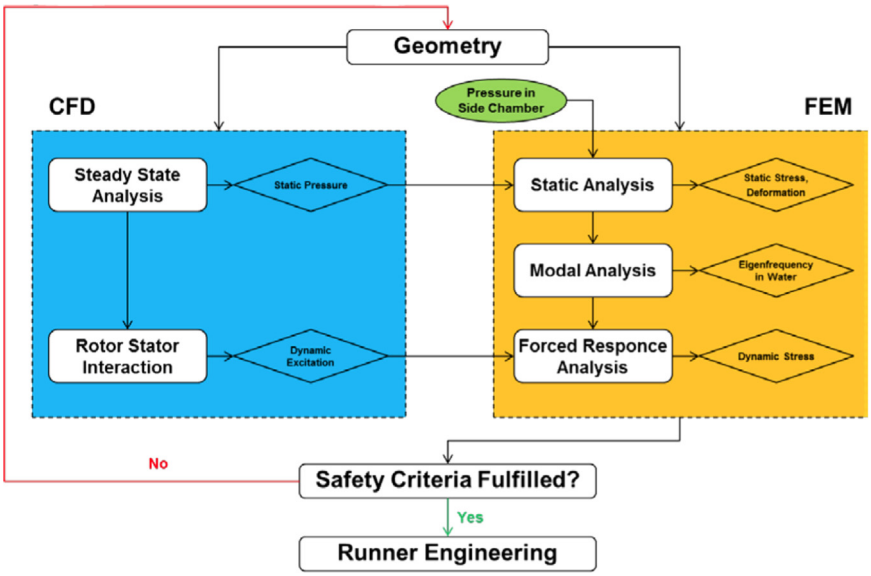


Fig. 10. Runner safety check procedure.

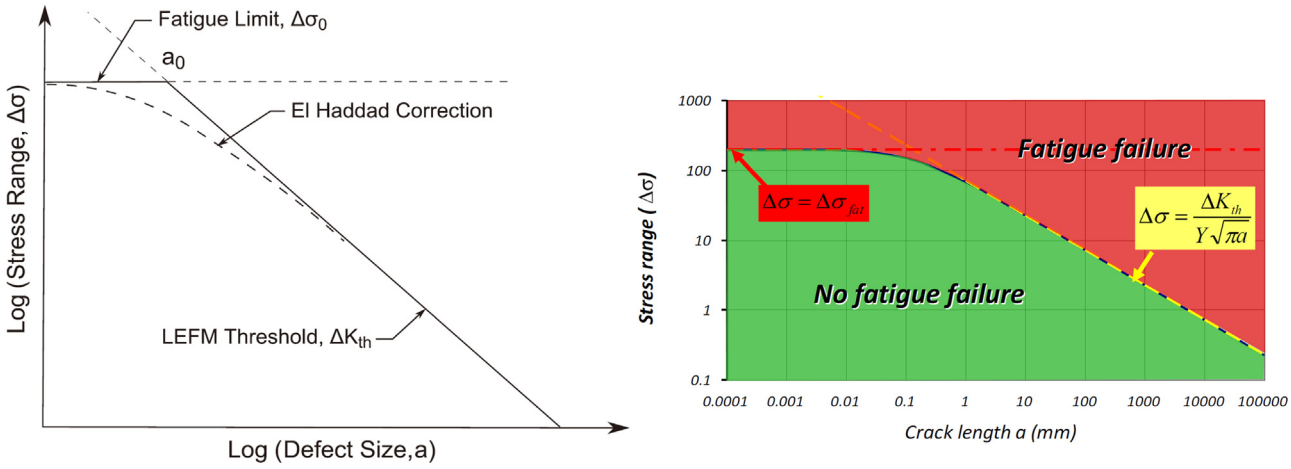


Fig. 11. Schematic Kitagawa diagram (left). Example (right).

Mixed mode crack growth behavior is difficult to be evaluated by experimental tests or experimentally inaccessible. Because it needs to test a large number of materials and structural components in a very short time. This implies the benefit of using numerical computational methods to estimate the 3D mixed mode crack propagation. The accuracy of the crack propagation estimate obtained by applying methods in this paper are still under thorough investigation. However, the work is a field of scientific interest and to provide a collection of references and a specific case for further research. And it is truly a very useful tool and will become a standard analysis technique in the future [90].

4.3. *Kitagawa–Takahashi diagram*

In some cases of large Francis turbines, there is a dilemma: if the HCF onset has occurred, a longer time between inspections leads to longer cracks to be repaired, and, at the same time, if the component is inspected before the HCF onset, additional downtime and maintenance costs incur with limited information on the state of the structure because the detectable flaw size has not been reached [91]. Gagnon et al. proposed to move away from the typical fatigue limits, as commonly seen in SN and critical crack lengths which do not adequately reflect this dilemma, in favor of a limit state directly related to the HCF onset [89]. Japanese researchers Kitagawa and Takahashi [92] first introduced the Kitagawa–Takahashi diagram to illustrate how the ΔK_{th} (fatigue crack growth threshold) tends to decrease when crack length decreases. Plotting their results of short fatigue crack growth testing on a stress range ($\Delta\sigma$) versus crack length (a), see Fig. 11·[75]. The plateau corresponds to the fatigue limit ($\Delta\sigma_0$) determined by *S–N* curves. This Kitagawa–Takahashi diagram illustrates how classical approaches based on *S–N* curves can be linked to damage tolerance approaches. It also shows what material properties have to be studied to correctly assess fatigue behavior of structures: the fatigue resistance ($\Delta\sigma_0$) and ΔK_{th}. More researchers extended this diagram to include other physical parameters such as notch effect [93], multi-axial criteria [94], and residual stresses [95,96]. A fatigue reliability model for hydroelectric turbine runners is proposed by Gagnon et al. [89]. In their model, reliability is defined as the probability of not exceeding a threshold above which HCF contribute to crack propagation. In the context of combined LCF–HCF loading, the Kitagawa diagram is used as the limit state threshold.

This two-dimensional diagram as shown in Fig. 11 (left) is established that the threshold which differentiates a propagating from a non-propagating crack [92], combined with the correction factor developed by El Haddad et al. [97], which accounts for short crack propagation. The limit state is characterized by at least two thresholds; the first threshold is the stress intensity range for crack growth ΔK_{th} as defined by Linear Elastic Fracture Mechanics

(LEFM), and the second threshold is the fatigue limit, which usually corresponds to 10^7 stress cycles for materials exhibiting an endurance limit.

The limit formed by the LEFM threshold is obtained from the stress intensity factor solution defined as follows:

$$\Delta K = \Delta\sigma\sqrt{\pi a}Y(a) \tag{2}$$

where ΔK is the stress intensity factor, $\Delta\sigma$ is the stress cycle range, a is the crack length and $Y(a)$ is the stress intensity correction factor for a given geometry. The limit state equation is obtained by:

replacing ΔK by the LEFM threshold ΔK_{th} in Eq. (2), which is rewritten as follows:

$$\Delta\sigma_{th} = \frac{\Delta K_{th}}{\sqrt{\pi a}Y(a)} \tag{3}$$

To capture short crack growth, El Haddad et al. [97] proposed to asymptotically match the limits defined by LEFM and the fatigue limit $\Delta\sigma_0$ using the reference crack length a_0 as a correction factor. The correction factor a_0 is added to the crack length a in Eq. (3) to obtain:

$$\Delta\sigma_{th} = \frac{\Delta K_{th}}{\sqrt{\pi(a+a_0)}Y(a+a_0)} \tag{4}$$

where the constant a_0 represents the transition between both limits, and is obtained by solving the following equation:

$$a_0 = \frac{1}{\pi}\left(\frac{\Delta K_{th}}{\Delta\sigma_0 Y(a_0)}\right) \tag{5}$$

For a structure like a hydroelectric Francis runner, two types of material flaws are usually considered critical: surface flaws and near-surface embedded flaws. Long cracks will grow if subjected to a stress intensity factor range ΔK which is greater than the long crack threshold value, ΔK_{th}. For short cracks, however ΔK levels are lower than this, provided that the stress amplitude is high enough. Specifically, the stress amplitude $\Delta\sigma$ must be greater than the fatigue limit, $\Delta\sigma_0$. The following is a simple fatigue strength prediction case [98].

These two curves are plotted in Fig. 12 (left) and together constitute a threshold condition for short and long crack growth. (Fig. 12 (left) is referred to as a 'modified' Kitagawa–Takahashi diagram, as the vertical axis is stress intensity factor rather than stress.) If the ΔK value as a function of crack length for a crack growing away from a notch at a particular load level, it is noted that there are three possible responses in this diagram. If the load is high enough, the curve will always be above the threshold (curve A) and failure will result. For a lower load (curve B), there will be no failure unless there is a pre-existing flaw of at least a_1 in length. For an intermediate load (curve C), the crack may propagate initially before falling below the threshold again and arresting. These observations may be used as a basis for determining a fatigue threshold for the notched component. For a specific notch size, ΔK is calculated as a function of crack length for a crack growing away from the notch. Then scale the resulting curve by adjusting the remote load until the ΔK curve is just tangent to the threshold. This load defines the threshold for the particular notch size. The procedure may then be repeated for different notch sizes to produce design curves as shown in Fig. 12 (right). Here, the predicted fatigue limit has been plotted as a function of notch depth b and notch root radius ρ for three different geometries, $\rho/b = 1.0$, 0.2, and 0.05. The vertical axis is the damaged fatigue strength as a percentage of the undamaged value. The predicted fatigue strengths is defined by Eq. (6). It will be seen that notch depth has a significant effect on fatigue performance. Small notches lead to higher fatigue strengths than large ones.

$$\frac{\Delta\sigma}{\Delta\sigma_0} = \frac{\Delta K}{\Delta K_0}\frac{\sqrt{a_0}}{\sqrt{a}} \tag{6}$$

4.4. A quick case on impact of startup scheme on Francis runner life expectancy

This case investigated life expectancy of Francis runner in two startup schemes using in situ measurements. Through this case, it presented the crack growth analysis method as well as the influence of one transient event. More details of this case could be seen in reference [58].

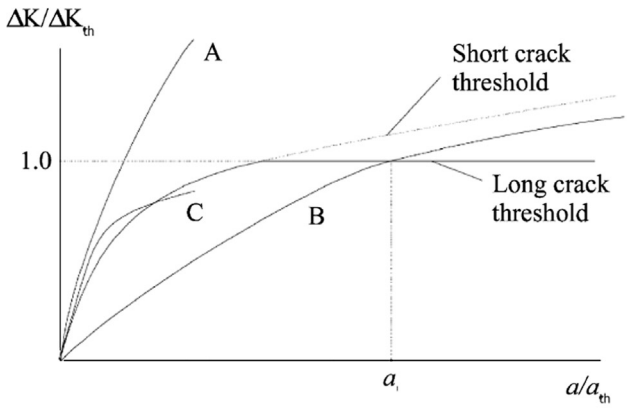
The crack growth calculation will be done in accordance with the fracture mechanic approach presented in the British Standard BS7910:2005. For each crack growth increment, the crack growth rate and stress intensity factor will be calculated with the following equations:

$$\frac{da}{dN} = C\Delta K^m \text{ for} \Delta K > \Delta K_{th}$$

In situ measurements made by IREQ (Institut de recherche d'Hydro-Québec) in summer 2002 at the Beauharnois hydroelectric plant in Québec Canada, have been used to obtain the response spectrum of two startup schemes. Only the location where maximum stress is expected will be used for this study. The chosen location is presented in Fig. 13.

The guide vane opening, the resulting time signal and the crack growth results for each startup scheme are presented in Fig. 14.

This case has shown that the optimization of the startup sequence could have a significant impact on life expectancy. Notice that only two different startup schemes have been evaluated in



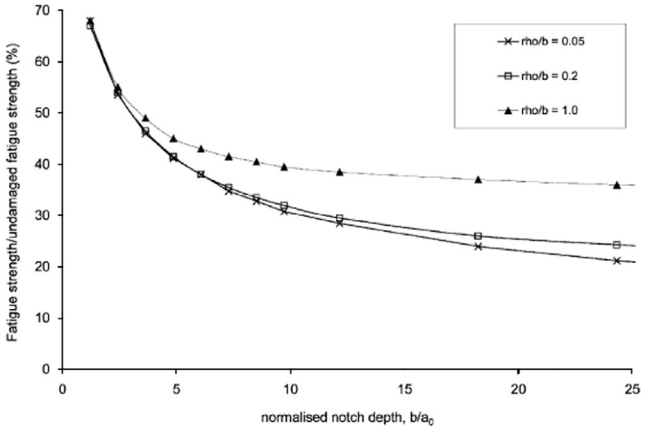


Fig. 12. Short crack arrest in the modified Kitagawa–Takahashi diagram (left); Predicted fatigue strengths ($\Delta\sigma/\Delta\sigma_0$) as a function of notch depth (b/a_0) for different notch root radii (ρ/b).

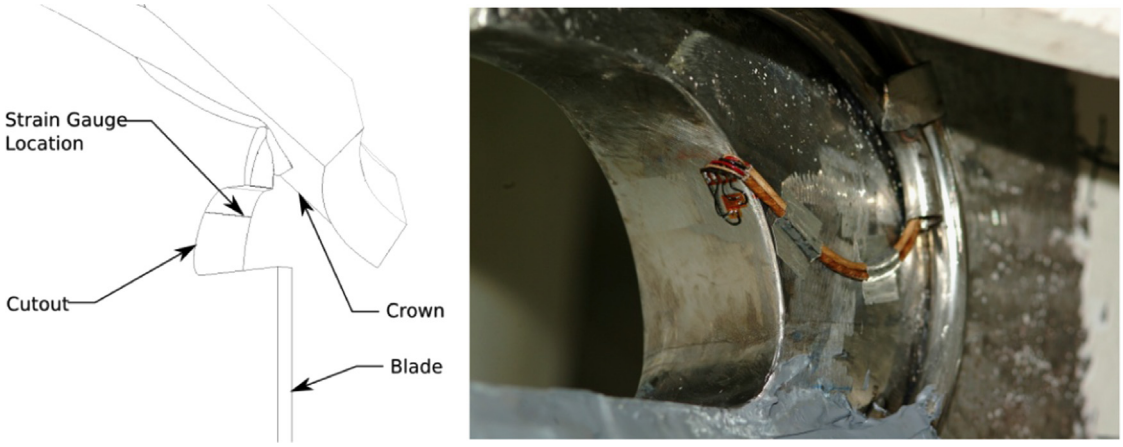


Fig. 13. Strain gauge location of the turbine runner in situ measurements; (left) Drawing; (right) Picture.

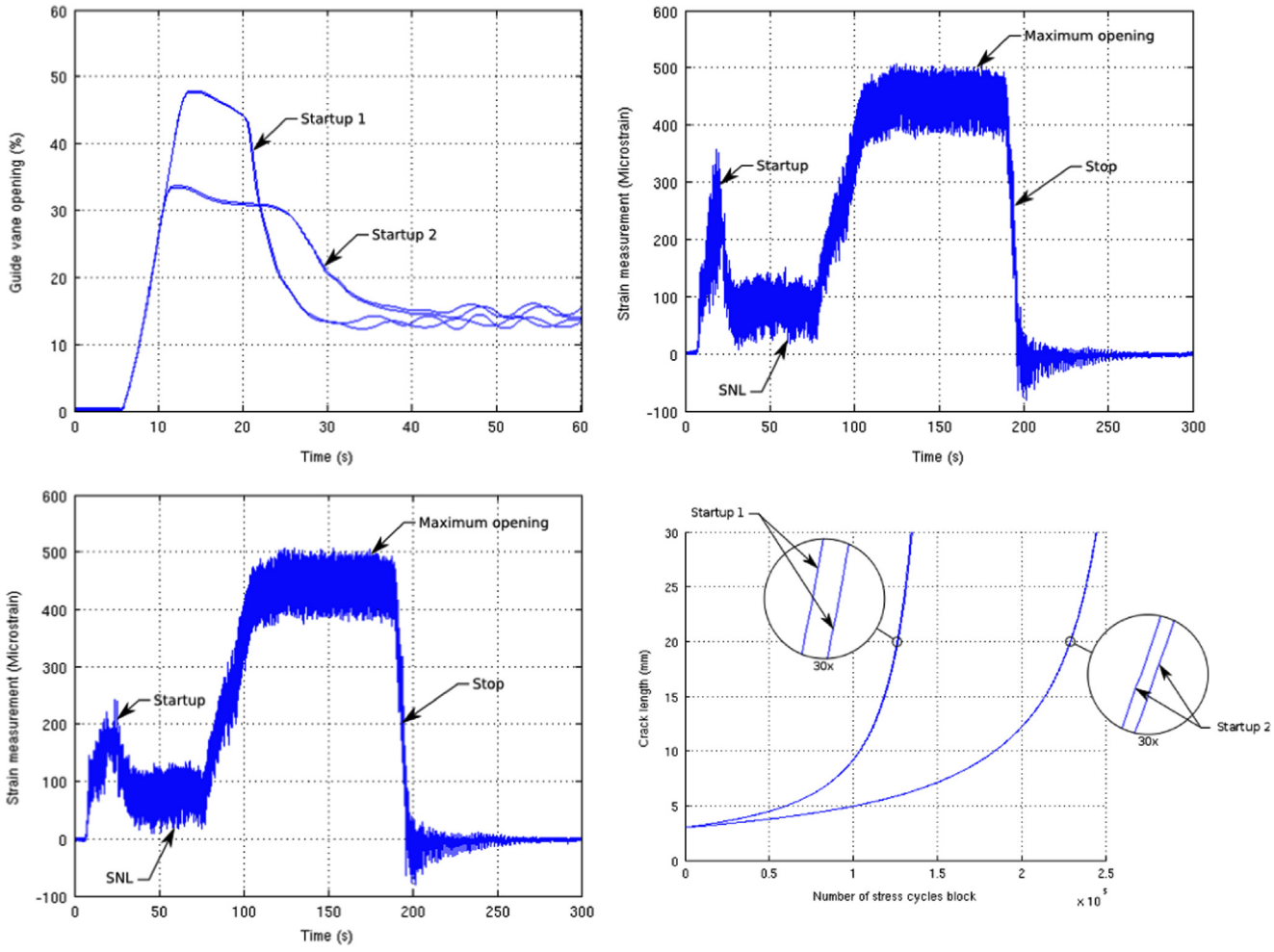


Fig. 14. The guide vane opening, the resulting time signal and the crack growth results for each startup; (upper right) Startup type 1; (lower left) Startup type 2.

this case and startup with even lower damage should be expected if more measurement were available. However, it does not expect every Francis runners to have the same behavior. Startup schemes should be analyzed on a case by case basis. Currently, there are only a few Francis turbines for which strain measurements are available. In the future, more field measurements will be required to develop a better understanding of the relationship between the guide vanes opening rate, the hydraulic force generated and the strain during runner startup.

5. Other main failure mechanisms

5.1. Cavitation

The water enters hydro turbines is subjected to changes in pressure and velocity. Such variations may result in changes in flow characteristics with consequences on turbine performance and useful life. Cavitation may occur near the fast moving blades of hydro-turbines or near the exit of the turbine where there are large differences between

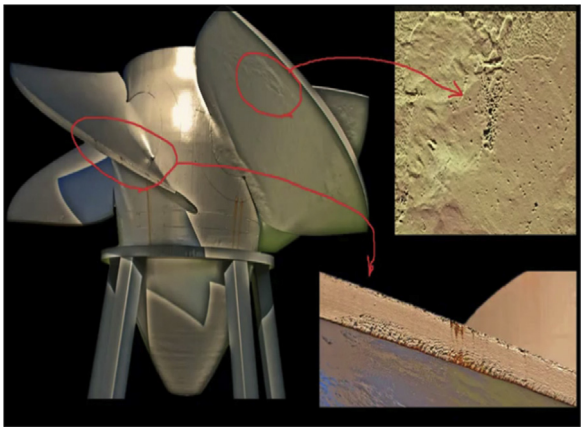


Fig. 15. Cavitation pitting damage occurs as a result of repeated fluid impact on the surface of a Kaplan turbine [98].

the static and dynamic components of fluid pressure. Dorji and Ghomashchi [99] gave a very good literature review on cavitation failure mechanism on hydro turbines. Repeated formation and collapse of vapor bubbles during the fluid flow deteriorates the surface of the machine components due to pitting action [100], as shown in Fig. 15. Simoneau [101] found that cavitation causes surface penetration damage of up to 10 mm per year to critical components such as impellors, turbine blades, and casings. Design profile of the turbine and the frequent change in the operating condition of the plant to meet various load requirements are two main causes of cavitation in hydro turbine [102]. There are four typeset of cavitation detected can damage the Francis turbines: leading edge cavitation, trailing edge cavitation, draft tube swirl and inlet blade vortex cavitation. The detailed effect mechanism was discussed by Kumar [103].

5.2. Erosion

Normally, erosion won't cause the turbine failure in a short-term. The harm extent of erosion is not as severe as fatigue failure, but it has a significant effect on the performance degeneration. In addition to this reduced performance, the turbine component may break down during any time of its service life causing danger to operational and maintenance crews [99]. For impulse turbines, wear on needle and nozzle would result in a decay of efficiency and possibly cavitation. In worn bucket, the boundary layer is thickened and disturbed due to an increased waviness of the surfaces [104]. And for reaction turbines, the performance of a turbine is destined to degrade due to various reasons as years go by. These factors include metal loss (cavitation, erosion, and corrosion), opening of runner seal, opening of guide vanes clearances, and increasing surface roughness. Erosive wear due to high content of abrasive material during monsoon and cavitation is the very important one [105]. The erosive wear of turbine and its components (e.g. guide vanes, disk and seal rings) in hydropower plants commonly occurs as a result of the flow of high velocity and impingement of abrasive sediments on the surface of the turbines [106]. Neopane et al. [107] evaluated the effect of sand particles on the turbines, and indicated that the erosive wear of turbine and its components is directly proportional to the sediment size and its mineral content. Similar results were also obtained by Kjølle [100] where in the erosion rate is directly determined by the sediment types and their characteristics properties such as shape, size and quantities.

5.3. Ingested bodies

Hydro turbines suck in large amounts of water that can transport small particles, such as sand, and large bodies, such as large stones and logs. The ingestion of large bodies in turbines can produce blockage in the runner that generates large, unbalanced forces, see Fig. 16. A body carried in the water will cause direct damage through impact or erosion and indirect damage through blockage. In the meantime blockage in the distributor can change the amplitude and uniformity of the pressure pulsations. Mechanical damage may also be generated. In all cases, the efficiency of the turbine is reduced and the forces on the runner and the vibration levels in the machine are greatly amplified, which reduces the expected life of bearings and seals and even produces considerable damage. Egusquiza et al. [108] presented an overview of the damage caused by the ingestion of external bodies in hydro turbines.

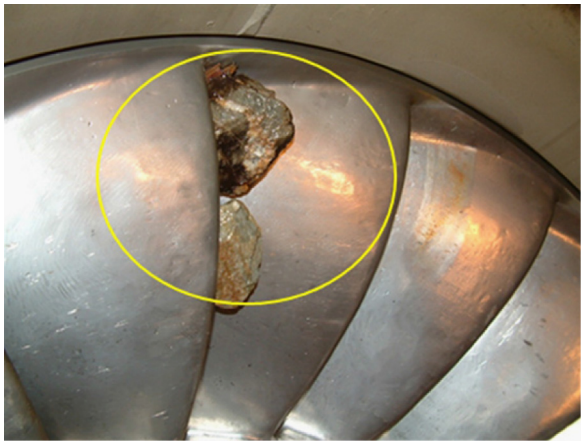


Fig. 16. Typical runner blockage.

6. Conclusions

This paper has carried out a review on fatigue damage mechanism of hydro turbines with particular emphasis on the latest discoveries in various operating conditions and developments in fatigue life prediction methods. All the factors which affect the fatigue life of hydro turbines are grouped into four categories: material, structure, loading, and operating conditions. The effects of these factors on fatigue behavior have also been addressed. In order to reveal the complete failure mechanisms in hydro turbines, other types of failure modes, such as cavitation, erosion and ingested bodies, are also introduced briefly.

Acknowledgments

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